

KAZI SHAIKAT HOSSAIN

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PROFESSIONAL SUMMARY

Recent B.Sc. graduate in Electrical and Electronic Engineering (GPA: 3.87/4.00) with undergraduate research experience in semiconductor device simulation, TCAD modeling, and high-k dielectric MOSFET optimization. Experienced with Silvaco TCAD, COMSOL Multiphysics, MATLAB, and Cadence. Strong background in nanoscale MOSFET characterization, technical writing, and analytical problem solving, complemented by several years of private tutoring and mentoring.

EDUCATION

B.Sc. in Electrical & Electronic Engineering

2026

Dhaka International University

GPA: 3.87 / 4.00

Relevant Coursework: Semiconductor Devices, Microelectronics, Digital Electronics, Power Electronics, Digital Signal Processing, Control Systems.

HSC

2020

Govt Sayed Hatem Ali College

GPA: 5.00 / 5.00 (Golden GPA5)

SSC

2018

Palordi Model School & College

GPA: 5.00 / 5.00 (Golden GPA5)

THESIS & RESEARCH

- Simulated 50 nm planar bulk nMOSFETs using Silvaco TCAD Atlas.
- Compared Si, 4H-SiC, and Ge channel materials with SiO₂/HfO₂ gate dielectrics.
- Investigated oxide thickness and metal work-function effects on device performance.
- Extracted Ion, I_{off}, Ion/I_{off}, threshold voltage, subthreshold swing, gate leakage, and thermal characteristics.
- Applied SRH, Auger, Fermi-Dirac, field mobility, and band-to-band tunneling physical models.
- Prepared undergraduate thesis and research manuscript.

Thesis:

Comparative Analysis of 50 nm Si, 4H-SiC and Ge MOSFETs with SiO₂ and HfO₂ Gate Dielectrics in Different Oxide Thickness.

Research Paper (Publications):

Comparative Performance Analysis of Si and SiC MOSFETs Using HfO₂ High-k Gate Dielectric and Different Metal Work Functions at Different Oxide Thicknesses (conference manuscript: Accepted- ECCT-2026).

ENGINEERING PROJECTS

- **MOSFET Device Simulation:** Designed and analyzed nanoscale MOSFET structures using Silvaco TCAD.
- **Smart Irrigation System:** Arduino-based automatic irrigation using soil moisture sensing.
- **DC-DC Boost Converter:** Designed and analyzed voltage step-up converter for power electronics.

EXPERIENCE

Private Tutor (Self-Employed)

~5 Years

- Tutored SSC and HSC students in Mathematics and Physics.
- Developed personalized lesson plans and assessments.
- Improved communication, mentoring, and time-management skills.
- Managed multiple students and schedules independently.

TECHNICAL SKILLS

Simulation: Silvaco TCAD Atlas, COMSOL Multiphysics, Cadence (Basic)
Programming: C, Arduino, Basic Verilog
Engineering: Semiconductor Devices, MOSFET Modeling, Device Physics, Circuit Analysis, Power Electronics, Digital Electronics
Software: CISCO Packet Tracer, Overleaf, Mendeley, MATLAB, Microsoft Office, Microsoft PowerPoint

LANGUAGES

Bengali: Native | English: Professional Working Proficiency